Bihar Engineering University, Patna

B.Tech 7th Semester Examination 2025

Course: B.Tech | *Time: 03 Hours*
Code: 105713 | *Subject: Cyber Security* | *Full Marks: 70*

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 ***Choose the correct option / answer the following (Any seven question only):*** **[2 x 7 = 14]**

(a) Which phase comes first in ethical hacking?
(i) Gaining access (iii) Reconnaissance
(ii) Scanning (iv) Maintaining access

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(b) Which cryptographic technique provides non-repudiation?
(i) Encryption (iii) Hashing
(ii) Digital signature (iv) Firewall

(c) Which technique captures packets for analysis?
(i) Encryption (iii) Hashing
(ii) Firewalling (iv) Packet sniffing

(d) Which is a major cause of security vulnerability?
(i) Strong passwords (iii) Firewall
(ii) Weak authentication (iv) Encryption

(e) Which malware hides attacker presence?
(i) Rootkit (iii) Worm
(ii) Virus (iv) Trojan

(f) Which is a major IoT security challenge?
(i) Large storage (iii) Fast processors
(ii) Weak security controls (iv) Large memory

(g) Which security mechanism verifies identity first?
(i) Authorization (iii) Encryption
(ii) Authentication (iv) Firewall

(h) OWASP is mainly concerned with
(i) Hardware security (iii) Web application security
(ii) Network security (iv) Physical security

(i) Which malware provides attacker remote control?
(i) Adware (iii) Bot/Zombie
(ii) Spyware (iv) Antivirus

(j) Hash functions are used to ensure
(i) Confidentiality (iii) Availability
(ii) Integrity (iv) Authorization

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Q.2 (a) You receive an email from an unknown sender claiming to be from your bank. The email says that your bank account will be blocked unless you verify your details immediately. The email asks you to click a link and enter your username, password, and ATM PIN. [7]

You notice that the email address is not the official bank email, but it looks similar. Answer the following:

(i) What type of cyber-attack is this
(ii) Explain how the attacker is trying to trick the user
(iii) What are the risks if the user enters their details
(iv) What should the user do after receiving such an email
(v) Suggest two preventive measures to avoid such attacks

(b) Explain the CIA triad in cyber security. Describe Confidentiality, Integrity, and Availability with suitable examples. [7]

Q.3 (a) Explain the process of encryption and decryption. Describe how encryption protects data during transmission with examples. [7]

(b) Explain symmetric key cryptography and asymmetric key cryptography. Compare their advantages and limitations with examples. [7]

Q.4 (a) Define digital signature. Explain its working and importance in ensuring data authenticity and integrity. [7]

(b) What is the role of Pretty Good Privacy in securing email communication? [7]

Q.5 (a) What is DDoS attack? Explain how to protect from DDoS attack? [7]

(b) Define packet sniffing. Explain how packet sniffing works and describe its role in network security analysis. [7]

Q.6 (a) Define cyber security vulnerabilities. Explain different types of vulnerabilities such as software vulnerability, network vulnerability, and human vulnerability with examples. [7]

(b) Explain authentication, authorization and access control. Compare their roles in cyber security with examples. [7]

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Q.7 (a) Define malware. Explain different types of malware such as virus, worm, Trojan, ransomware, spyware, and rootkit with examples. [7]

(b) Explain different methods used to protect systems from malware attacks using antivirus, OS hardening, and security tools. [7]

Q.8 (a) Define mobile computing. Explain different security threats in mobile devices such as malware, data theft, and unauthorized access. Describe methods to secure mobile devices. [7]

(b) What is Secure Socket Layer? Explain its significance in web security and provide a detailed description of the various SSL protocols, including the phases of the Handshake Protocol."? [7]

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Q.9 Write short note on *any two* of the following: [7 x 2 = 14]

(a) Steganography
(b) Types of Firewalls
(c) National Cyber Security Policy 2013 of India
(d) Cyber Forensics Investigation Steps

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